

PRANATI JANA (She/Her)

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EDUCATION

Ph.D | Indian Institute of Technology (IIT) Madras, India *Sep 2020 - Jan 2026*
Experimental High Energy Physics, CMS Collaboration, CERN
Supervisor: Prof. Prafulla Kumar Behera

M.Sc | National Institute of Technology Durgapur, India | CGPA: 9.11/10 *2018-2020*
Major: Physics (Awarded with “First Class”)

B.Sc | Asutosh College, Calcutta University, India | Marks: 76.2% *2015-2018*
Major: Physics, Minor: Mathematics, Chemistry (Awarded with “First Class”)

RESEARCH EXPERIENCE

Fermilab LPC G&V programme | CMS outer tracker upgrade *Sep 2025 - Feb 2026*

- Gained experience in sensor testing and software development, a machine learning project.

Project 1 | Measurements of the light-by-light scattering and the Breit–Wheeler processes, and searches for axion-like particles in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV at CMS [1]

- From this study, I gained in-depth knowledge about low- p_T photons and electrons (identification, reconstruction, e/γ regression, etc.) as well as a deeper understanding of the performance of calorimeters.

Project 2 | Measurement of the tau $g - 2$ factor in the ultraperipheral PbPb collisions recorded by the CMS experiment at $\sqrt{s_{NN}} = 5.02$ TeV [2]

- In this analysis my work primarily focuses on the electron+muon decay channel.

Project 3 | Radiation hardness study of MALTA2 monolithic CMOS imaging sensors on Czochralski substrate [3]

- Participated in the 2023 test beam campaign, where I tested MALTA2 chips with different configurations and irradiation doses, and analyzed the data, gaining valuable hands-on experience in sensor characterization and performance studies, including tracking efficiency, cluster size calculation as well as system tests, including threshold and noise studies.

Project 4 (Authorship task)| CMS luminosity calibration of XeXe collisions at 5.44 [4]

Master’s | Photodetection performance of $\text{TiO}_2\text{-CuO}$ based heterostructure film *2019-2020*

- Gained experience in device fabrication using doctor blade, thermal evaporation.

PUBLICATIONS

- [1] CMS Collaboration, “Measurement of light-by-light scattering and the breit-wheeler process, and search for axion-like particles in ultraperipheral pbpb collisions at $\sqrt{s_{NN}} = 5.02$ tev,” *JHEP*, vol. 2025, no. 8, p. 6, 2025. DOI: 10.1007/JHEP08(2025)006.
- [2] “Measurement of the tau $g-2$ factor in ultraperipheral PbPb collisions recorded by the cms experiment,” CERN, Geneva, Tech. Rep., 2024. [Online]. Available: <http://cds.cern.ch/record/2912969>.

- [3] “Radiation hardness of MALTA2 monolithic CMOS imaging sensors on Czochralski substrates,” *Eur. Phys. J.*, vol. C84, p. 251, 2024. DOI: <https://doi.org/10.1140/epjc/s10052-024-12601-3>.
- [4] CMS Collaboration, “Luminosity calibration of 2017 XeXe collisions at 5.44 TeV,” CERN-CMS-DP-2024-015. [Online]. Available: <https://cds.cern.ch/record/2891266?ln=en>.

CONFERENCES AND WORKSHOPS

- “Photon–Photon Physics in Ultra-Peripheral PbPb Collisions at CMS and my activity on Silicon Detector R&D for future detectors” | LPC Physics Forum, Fermilab 2026
- “Exclusive photon and lepton production in UPC PbPb @5.02TeV at CMS” | Speaker at PHANC’25, Puri India 2025
- “Radiation hardness study of MALTA2 sensor” | Speaker at DAE BRNS symposium 2024, IIT BHU, Varanasi, India 2024
- “Measurement of the tau g-2 factor in ultraperipheral PbPb collisions recorded by the CMS experiment” | Speaker at DAE BRNS symposium 2024, IIT BHU, Varanasi, India 2024
- “Exclusive photon and lepton production in UPC PbPb @5.02TeV at CMS” | Speaker at (PPC 2024), IIT Hyderabad, India 2024
- “Measurement of Breit-Wheeler process, light-by-light scattering and searches for axion-like particles in UPC PbPb @ 5.02TeV at CMS” | Speaker at HardProbes 2024, Nagasaki, Japan 2024
- Participated in “Experimental High Energy Particle Physics School on software development school” at TIFR Mumbai, India 2024
- “Exclusive production on UPC PbPb collisions” | Speaker at Heavy Ion Physics Workshop at ECT*, Trento, Italy 2023
- “Measurement of the tau g-2 factor in ultraperipheral PbPb collisions at CMS experiment” | Speaker at In-House Symposium at Department of Physics, IIT Madras, India 2023
- “Observation of $\gamma\gamma \rightarrow \tau\tau$ production in PbPb collisions with the CMS experiment” | Speaker at DAE-BRNS High Energy Physics (HEP) Symposium 2022, IISER Mohali, India 2022
- “Measurement of the tau g-2 factor in ultraperipheral PbPb collisions at CMS experiment using electron+muon decay channel” | Speaker at AEP SHEP2022, at Pyeongchang (Korea, Republic Of) 2022
- Participated in summer School on “Quantum Information and Technology” (QIQT2019) at IISER Kolkata, India 2019

TEACHING EXPERIENCES

Facilitator	CMSDAS2026 at LPC Fermilab
Teaching Assistant	Laboratory for Physical Property Measurement and Sensor Element; Characteristics of Functional Materials; Physics Lab III (PG); Physics Lab for Engineering Physics

SKILLS

Languages	Bengali (Native), English (Professionals), Hindi
Programming Languages	Python, C/C++, ROOT

EXTRA-CURRICULAR ACTIVITIES

- Volunteer: Tribal Student Campus Visit Program, Research Scholar Immersion Programme, Beach Cleaning Drives, Campus Bird Count by WWF etc.